

Test 1-2

$$1. \quad \frac{4}{5} \left(10 - \frac{5}{4} \right) = \frac{4}{5} \left(\frac{40}{4} - \frac{5}{4} \right)$$

$$= \frac{4}{5} \times \frac{35}{4} = \underline{7}$$

$$2. a) 64^{1/3} = \underline{4}$$

$$b) (-27)^{1/3} = \underline{-3}$$

$$c) \left(-\frac{1}{27} \right)^{1/3} = \underline{-\frac{1}{3}}$$

$$3. \quad (6x - 3)(x^2 - x + 1) = \underline{6x^3 - 6x^2 + 6x - 3}$$

$$4. \quad \frac{x-9}{x^2-81} = \frac{x-9}{x^2-9^2}$$

$$= \frac{x-9}{x-9} = \underline{1}$$

$$5. \quad \cancel{17x} \quad |7x| = 4$$

$$7x = 4$$

$$x = \underline{\frac{4}{7}}$$

$$6. \quad |3x + 5| < 7$$

$$3x + 5 < 7$$

$$3x < 2$$

$$x < \underline{\frac{2}{3}}$$

$$\underline{(-\infty, \frac{2}{3})}$$

$$7. -4x + y = 2$$

$$y = -4x + 2$$

$$m = \underline{-4}$$

$$c = \underline{(0, 2)}$$

$$8. f(x) = x^3 + 2x$$

$$f(-2) = -8 - 4 = \underline{-12}$$

$$f(-1) = -1 - 2 = \underline{-3}$$

$$f(0) = \underline{0}$$

9. I is a function

$$10. f(x) = \sqrt{x-6}$$

$$x-6 \geq 0 \Rightarrow x \geq 6$$

$$\text{Dom: } [6, \infty)$$

$$\text{Range: } [0, \infty)$$

$$11. x^2 - 4x + 5 = (x-2)^2 + 1$$

$$\text{Vertex: } (2, 1)$$

$$x\text{-int: } (2, 0)$$

$$y\text{-int: } \cancel{2} (0, 5)$$

$$\text{Dom: } (-\infty, \infty)$$

$$\text{Range: } [1, \infty)$$

$$12. f(x) = 2^x \text{ goes up}$$

$$g(x) = 2^{-x} \text{ goes } \cancel{\text{up}} \text{ down}$$

Bottom-right graph

$$13. \quad t = \frac{3\pi}{2}$$

$$\underline{(0, -1)}$$

$$14. \quad \frac{7\pi}{6} = \frac{7}{6} (180) = \underline{210^\circ}$$

$$15. \quad (-4, \frac{4\pi}{3})$$

$$x = t \cos \theta = (-4) \left(-\frac{1}{2}\right) = 2$$

$$y = t \sin \theta = (-4) \left(-\frac{\sqrt{3}}{2}\right) = 2\sqrt{3}$$

$$\underline{(2, 2\sqrt{3})}$$

$$16. \quad p = (2, 4) \quad q = (7, 3)$$

$$\langle 7-2, 3-4 \rangle = \underline{\langle 5, -1 \rangle}$$

$$17. \quad 12 - 19 = -7$$

$$5 - 12 = -7$$

$$-4 - 5 = -9$$

Not arithmetic

DNE